BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Parameters, Return Values and Scope

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How is a cell selected in a two-dimensional array?

2. What should a useful code comment explain?
   A) The programmer's password
   B) An unrelated topic
   C) A misleading instruction
   D) Why the code is needed

3. "The CPU's main purpose is permanent file storage." Correct this. Explain why.

4. A folder contains 20 images of 200 KB each. Using decimal units, give its total size in KB and MB.

Today's objectives

Explain the core ideas of parameters, return values and scope using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Parameter
A value passed to a subprogram when it is called.

Return value
A function sends a result back to the calling code.

Local scope
A local variable is accessible within its subprogram.

Global scope
A global variable is defined outside subprograms and can be accessed more widely, increasing risk of unintended changes.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Parameter
A value passed to a subprogram when it is called.

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A local variable is accessible within its subprogram.

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A global variable is defined outside subprograms and can be accessed more widely, increasing risk of unintended changes.

I Do - Worked model

Trace a function double(x) that returns x*2 when called with 7.

- The argument 7 is passed into parameter x.
- Inside the function x*2 gives 14.
- The function returns 14 to its caller.
- If printed, the output is 14.

We Do - Guided practice

1. Trace a function addTax(price) which returns price+2 for input 9.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the line using the global number of bits per byte.

```
global bitsPerByte = 8
function toBits(bytes)
totalBits = ...
return totalBits
endfunction
print(toBits(7))
```

You Do - Independent exam questions

Q2. Differentiate a function and a procedure. [2 marks]

Q3. A function returns x+5. What is returned for x=8? [1 mark]

Q4. Explain one benefit of local variables. [2 marks]

Exit Ticket - True or False?

A local variable is necessarily accessible from every subprogram.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Use two indexes: one for the row and one for the column.

2. D - Why the code is needed

3. It executes instructions to process data.

4. (a) 4,000 KB = 4 MB.

I Do - Explain the model

The argument 7 is passed into parameter x.

Inside the function x*2 gives 14.

The function returns 14 to its caller.

If printed, the output is 14.

We Do - Model answers

1. Return value 11.

Code First - Answer

```
global bitsPerByte = 8
function toBits(bytes)
totalBits = bytes * bitsPerByte
return totalBits
endfunction
print(toBits(7))
```

Expected output: 56

You Do - Self-mark

Q2: A function returns a value; a procedure performs an action and does not need to return a value.

Q3: 13.

Q4: They limit unintended interference between subprograms and make code easier to reason about.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.